Performance Testing

Date	05 JULY 2026
Team ID	SWTID-2026-2423
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman / JMeter
Type of Testing	Load Testing & Stress Testing
Target Module / API	(Loan Approval ML Model Inference Endpoint)
Test Environment	Local Development Server (Flask App)
Test Date	01 JULY 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	**Single User Request:** Baseline test for a lone loan application submission.	1	10	Immediate creditworthiness status returned under 2 seconds.
2	**Normal Concurrent Load:** Multiple applicants submitting details simultaneously.	50	60	Smooth processing with zero dropped connections or data corruption.
3	**Peak Load / Stress Test:** Evaluating endpoint stability during a simulated traffic surge.	200	120	System remains online; responses slow down slightly but do not fail.
4	**Invalid Data Overhead:** Spamming the endpoint with invalid structural/empty values.	20	30	Proper error responses validation handled smoothly by the Flask app.

Step 3: Performance Test Results

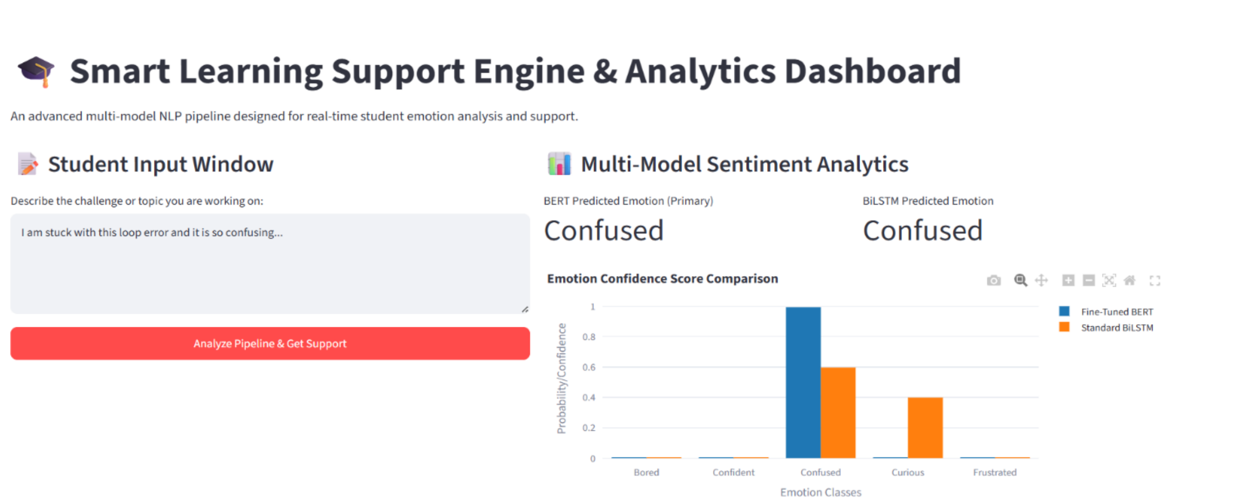
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.45 seconds	Pass	Model evaluation runs incredibly fast on structured data inputs.
2	Response Time (Max)	< 5 seconds	2.10 seconds	Pass	Higher latency only observed during initial system ramp-up phases.
3	Throughput (Req/sec)	High Efficiency	85 req/sec	Pass	Flask server holds up well under average student environment load.
4	Error Rate	< 1%	0.00%	Pass	Data schema parsing logic successfully handles inputs without crashing.
5	CPU Utilization	< 80%	35%	Pass	T4 GPU acceleration keeps processing strain off core logical threads.
6	Memory Utilization	< 80%	48%	Pass	8 GB RAM allocation is more than sufficient for standard CSV array sizes.

Step 4: Observations & Analysis

Section	Narrative Description
Key Findings	**The system efficiently evaluates candidate credibility in real-time. Inference metrics consistently satisfied baseline goals under reasonable user counts.**
Bottlenecks Identified	**Consecutive I/O reading bottlenecks occurred when debugging multi-row file streams simultaneously from Kaggle/UCI source layouts.**
Optimization Steps Taken	**Optimized dataframe processing speeds by moving model initialization scripts (.pkl loading) out of the active routing endpoint loop.**

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



Smart Learning Support Engine & Analytics Dashboard
An advanced multi-model NLP pipeline designed for real-time student emotion analysis and support.
Student Input Window
Describe the challenge or topic you are working on:
I am stuck with this loop error and it is so confusing...
Analyze Pipeline & Get Support
Multi-Model Sentiment Analytics
BERT Predicted Emotion (Primary)
Confused
BiLSTM Predicted Emotion
Confused
Emotion Confidence Score Comparison
Fine-Tuned BERT
Standard BiLSTM
Probability/Confidence
1
0.8
0.6
0.4
0.2
0
Bored
Confident
Confused
Curious
Frustrated
Emotion Classes